

Musical ‘Lick’ Modifier

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Background / Related Works

As a jazz musician, I often find myself repeating the same licks—short melodic phrases used during improvisation or as recurring motifs in a piece. This project aims to develop an application that enhances user-inputted MIDI files through algorithmic composition techniques, introducing variation and creativity into improvisation. There are two primary approaches to computer-generated music: self-learning systems and rule-based systems. The key distinction between them is that in rule-based systems, the coder manually defines musical rules and implements algorithms, whereas self-learning systems rely solely on training data with minimal human intervention. I’ve chosen to move forward with a more rule-based approach; however given the complexity of this project, a single model is unlikely to be sufficient. Instead, I intend to incorporate multiple models and techniques to handle both analysis of input and generation of output.

In his paper, John Wade Ulrich claims that the jazz artist has two main problems. The first is the creation of melodic material. The second is fitting that melodic material within a defined harmonic structure. Ulrich implements a non-grammatical approach to harmonic analysis, which is different from Winograd’s grammatical approach to tonal harmony in which he analyzes classical music. One of the major limitations that Ulrich cites on Winograd’s approach is that his implementation of will probabilistically pick the more simple of two possible chord progressions, which may not always be correct. I am interested in expanding on this work by expanding on the first problem, as I envision the user will input the chords, and the generated improvisation will have inspiration from the user-inputted MIDI file.

Goal and Objectives

I want this project to use rule-based composition instead of deep learning frameworks because it offers a more transparent, controllable, and interpretable approach to music generation. Unlike a self-learning system, which often operates as a "black box," I will clearly define rules that will allow users to understand how their input will be transformed. One of the major disadvantages of a self-learning system is that when it fails, it breaks spectacularly. Following a rule-based approach ensures that the

modifications remain musically coherent while still providing novel musical phrases. I believe this to be attainable because instead of creating entire compositions, we are specifying our program to reimagine a short phrase of music. Additionally, I want the tool to be as simple as possible for users - I aim to have as few inputs as possible and not overwhelm the user with potentially unpredictable AI-generated results. Ideally, our program will only take as input a chord progression and a jazz lick.

Mathematical Relevance

Music is inherently mathematical. The core of rhythm and harmony can be analyzed using group theory, set theory, and combinatorics. Additionally, mathematical composition is a natural application of advanced math. Brian Miller discusses how Georgia Tech's "Shimon" – a robot marimba player – uses hidden Markov models to model harmonic function. Numerous probabilistic models may be employed for both musical analysis and generation. Below, I outline one possible approach that may prove effective.

In terms of analysis, implementation of a grammar-based compression approach to analyze short segments of music may be sufficient. I plan to use context-free grammars to extract recurring patterns in a symbolic music sequence, allowing me to measure its structural complexity. By applying a well-defined compression algorithm, I aim to determine significant parts of the lick's structure to feed into the generating function.

In terms of generation, I plan to implement a distance pattern model to constrain a generative model. In his thesis paper, Jean-Francois Paiement defines a rhythm prediction model that used a distance pattern model to place constraints on a hidden Markov model to accurately predict rhythmic variation. There is untapped potential to implement such distance constraints on other predictive models. He also details using an Input Output Hidden Markov Model for Naumar Features (based on the Implication-Realization model of musical expectation) to predict and generate melodies given a chord progression and rhythm pattern. I would like to combine the two together to generate melody and rhythm simultaneously.

Timelines/Deliverables

I will divide development and research into two phases. Phase 1 will be the creation of our parser, and phase 2 will be the creation of our music generator.

1. Function that can parse the MIDI file, given a chord progression, and decide important information
 - a. Feb 7: decide on a language that has the ability to generate MIDI, write a lick with a chord progression

- b. Feb 14: implement context-free grammar framework
 - c. Feb 21: implement the compression algorithm for parsing
- 2. Function that can take the output from 1) and generate a related lick
 - a. Feb 28: Begin implementation of the generative model
 - b. Mar 7: Begin implementation on distance constraints
 - c. Mar 28: Have a working program that can take the lick written in 1a and generate different outputs
 - d. Apr 4: Have a working program that can generate based on different licks
 - e. Apr 11: Finish coding our application
- 3. Final Report/Demo
 - a. Feb 21: Begin implementing notes into a draft of the formal thesis paper
 - b. Apr 17: Finish writing first draft for review, create a video that demos the project, including folders of original licks and generated licks
 - c. Apr 21: Finish writing final report for final check, upload all code and generated content to GitHub

References

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